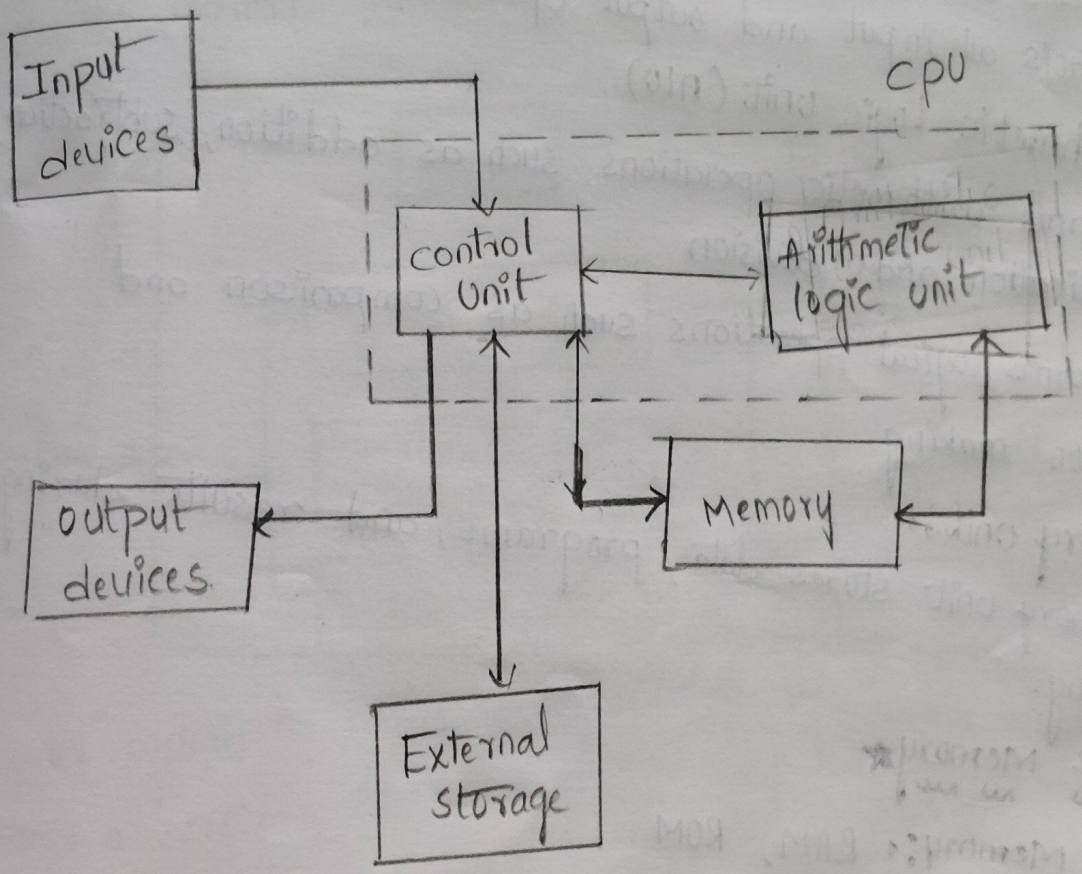
Subject Name: Digital design and computer organisation.
Subject code: BCS302.

(1) Explain the basic operational concepts of the computer with a neat diagram?

Ans:- A computer is an electronic machine that performs operations on data to generate meaningful information. The working of a computer follows the Input - process - output cycle.

Neat Block Diagram of a computer:-



* Input Unit:-

The input unit receives data and instructions from the user and converts them into a form suitable for processing.

Examples:
Keyboard
Mouse
Scanner
Microphone

* central processing unit (CPU)

The CPU is the main component responsible for processing data and controlling computer operations.

(a) control unit (CU)

→ fetches instructions from memory
→ Decodes and executes instructions
→ controls all input and output operations.

(b) Arithmetic Logic unit (ALU)

→ performs arithmetic operations such as addition, subtraction, multiplication, and division
→ performs logical operations such as comparison and decision making.

* Memory Unit:

The memory unit stores data, programs, and results during processing.

Types of Memory ★

primary Memory: RAM, ROM

Secondary Memory: Hard Disk, SSD, pen drive.

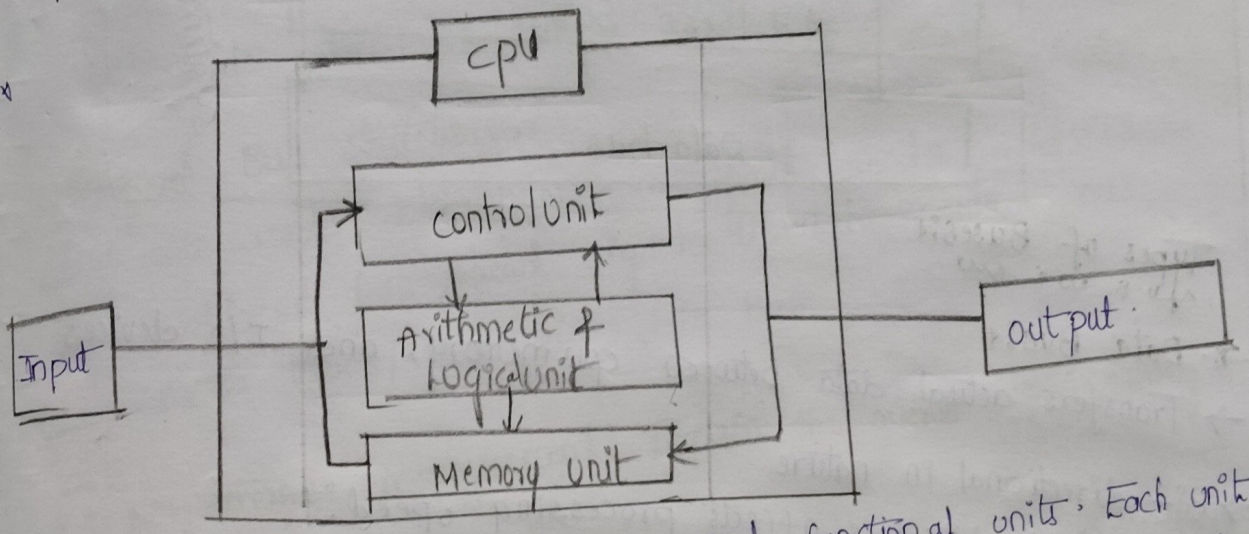
* output unit:-
The output unit displays the processed information in a human-readable form.

Examples:-
Monitor
printer
speakers.

Example of computer operation.
When two numbers are entered using the keyboard, they are stored in memory. The ALU processes the numbers and calculates the result, which is the displayed on the monitor..

(2) Explain the various functional units of computer with neat diagram?

Ans:-
working



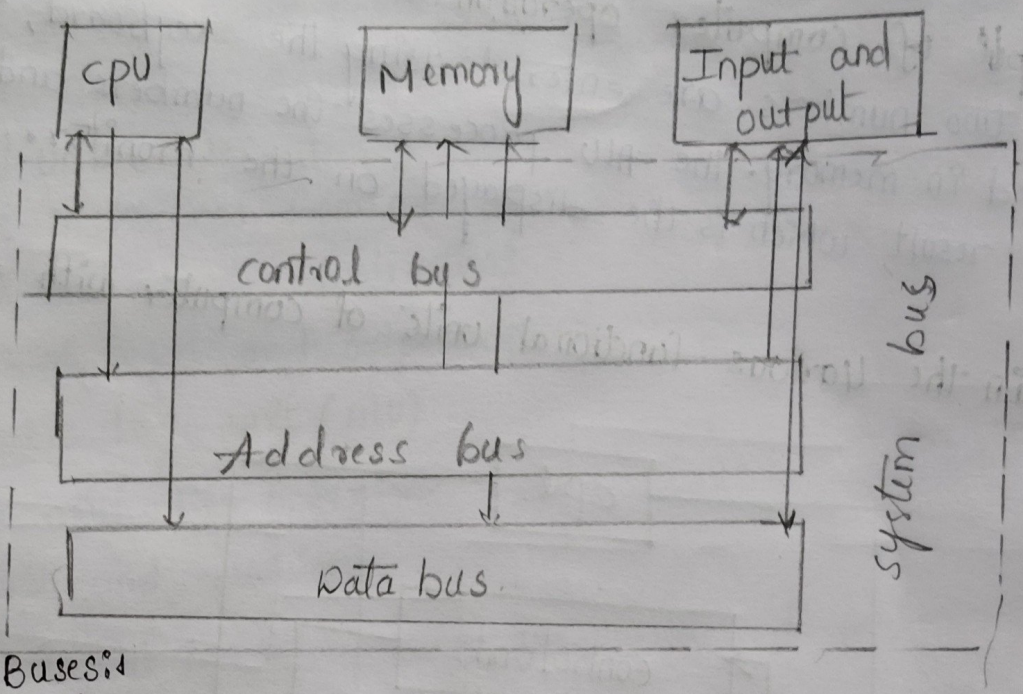
A computer system is made up of several functional units. Each unit performs a specific task and works together with other units to process data and produce useful information.

functional units of a computer:-
* Input unit
* central processing unit
* Memory unit
* output unit.

(3) Explain bus structure in computer?

Ans: A bus structure in a computer is a communication system that transfers data, address, and control signals between the cpu, memory, and input/output devices.

Neat diagram of bus structure:



Types of Buses:

* Data Bus:
→ Transfers actual data between cpu memory, and I/O devices
→ Bi-directional in nature
→ width of data bus affects processing speed.

Example:
Data read from memory is transferred to cpu through the data bus.

* Address Bus:
→ Carries address of memory or I/O location
→ Unidirectional (cpu to memory & I/O)
→ Determines maximum memory size.

Example:
CPU sends memory address through address bus to access data.

* Control Bus:
→ Carries control and timing signals
→ Controls read / write operations
→ Synchronizes data transfer.

Example:
Read or write signal is sent through control bus.

Working of Bus structure:
When CPU reads data from memory, it sends address through address bus, read signal through control bus, and data is transferred via data bus.

(4) What is performance measurement? Explain the overall spec rating of computer.

Ans: Performance measurement is a key concept in computer organization used to evaluate the efficiency, speed, and effectiveness of a computer system. It helps in comparing different systems and improving their design.

performance Measurement

performance measurement refers to calculating how fast a computer system executes programs.

The most widely used performance metric is execution time.

$$\text{performance} = \frac{1}{\text{Execution Time}}$$

other metrics include throughput, which measures the number of tasks completed per unit time.

cpu performance Equation

cpu Time = Instruction count x cpI x clock cycle Time.

Factors Affecting overall speed of a computer.

1. clock speed:
Measured in GHz; higher speed means faster instruction execution.
2. processor Architecture:
Multi-core cpus execute multiple programs simultaneously.
3. Instruction set & cpI
Lower cpI improves performance.
4. Cache Memory
Faster access than main memory, improves execution speed.
5. Main Memory (RAM)
More RAM supports better multitasking.
6. secondary storage
SSDs provide faster read / write operations than HDDs.

7. Bus structure
Faster data, address, and control buses improve system communication

8. operating system and Software.
Efficient os scheduling and optimized programs increase performance.

(5) Explain: i) processor clock
ii) clock rate
iii) performance equation.

Ans: In a computer system, the speed and performance of the processor are mainly determined by the processor clock, clock rate, and performance equation. These concepts help in understanding how fast a cpu executes instructions and how overall system performance is measured.

(i) processor clock:

Definition:
A processor clock is an electronic signal that synchronizes all operations inside the cpu. It acts like a timing signal that controls the execution of instructions in a step by step manner.

Explanation:
→ The processor clock generates a sequence of regular pulses.
→ Each pulse is called a clock cycle.
→ Every operation inside the cpu (fetch, decode, execute) occurs according to these clock cycles.
→ The clock ensures that all components of the cpu work in coordination.

Importance of processor clock

→ controls the speed of instruction execution

→ synchronizes CPU operations

→ Maintains proper data flow between CPU components

Example:

If a processor performs one basic operation per clock cycle, then more clock cycles mean more work can be done in less time.

(ii) Clock Rate:

Definition:-

Clock rate is the number of clock cycles generated by the processor clock in one second. It is also known as clock frequency.

Unit:

Measured in Hertz (Hz)

Modern processors use MHz or GHz.

Explanation:-

→ Higher clock rate means the CPU can perform more operations per second

→ clock rate determines how fast instructions are processed.

Example:

→ A processor with 1 GHz clock rate performs 1 billion clock cycles per second

→ A 3 GHz processor is generally faster than a 2 GHz processor.

Limitation:-

Higher clock rate alone does not guarantee better performance because:

→ Instruction complexity

→ CPI

→ processor architecture also matter

(iii) performance Equation

The performance equation relates execution time with Instruction count, CPI and clock cycle time. It helps in measuring cpu performance.

Instruction count (Ic):

Total no. of instructions executed

CPI: ~~Total number of instru~~

Average no. of clock cycles per instruction

Clock cycle Time:

Time taken for one clock cycle

Alternate form:

$$\text{CPU Time} = \frac{\text{Instruction count} \times \text{CPI}}{\text{clock rate}}$$

Importance of performance Equation

* Helps compare different processors

→ Identifies performance bottlenecks

→ Guides system optimization

Example:-

if

Instruction count = 1,000,000

CPI = 2

clock Rate = 2GHz.